## Exercises

**Exercise 8.1 · Prime or not?.** Are the following integers prime? 211; 221; 1,001; 257.

**Solution.** 211 is prime (course). $221 = 13 \times 17$. $1{,}001 = 7 \times 11 \times 13$. For 257: none of 2, 3, 5, 7, 11, 13 divides it and $17^2 = 289 > 257$: it is prime.

**Exercise 8.2 · Factorisation and divisors.** Factorise 720 into primes. How many positive divisors does it have? Are there many divisors of 720 that are perfect squares?

**Solution.** $720 = 2^4 \times 3^2 \times 5$. Number of divisors: $(4+1)(2+1)(1+1) = 30$. A square divisor has even exponents: $2^{\beta_1}3^{\beta_2}5^{\beta_3}$ with $\beta_1 \in \{0, 2, 4\}$, $\beta_2 \in \{0, 2\}$, $\beta_3 = 0$; there are 6 of them: 1, 4, 16, 9, 36, 144.

**Beyond the syllabus · Least common multiple.** The least positive common multiple of two nonzero integers, written lcm, is not part of the syllabus. It is used in the next exercise, which is not assessed.

**Exercise 8.3 · GCD and LCM by factorisation.** Factorise 1,260 and 1,050, then deduce their gcd and their least common multiple. Check that the product of the gcd and the lcm equals the product of the two numbers.

**Solution.** $1{,}260 = 2^2 \times 3^2 \times 5 \times 7$ and $1{,}050 = 2 \times 3 \times 5^2 \times 7$. GCD: $2 \times 3 \times 5 \times 7 = 210$; LCM: $2^2 \times 3^2 \times 5^2 \times 7 = 6{,}300$. And $210 \times 6{,}300 = 1{,}323{,}000 = 1{,}260 \times 1{,}050$.

**Exercise 8.4 · Fermat and remainders.**

1. Determine the remainder of $2^{2026}$ in the division by 13.
2. Prove that for every integer $n$, $n^{13} - n$ is divisible by 13.

**Solution.**

1. 13 is prime and does not divide 2: $2^{12} \equiv 1 \pmod{13}$. $2{,}026 = 12{\times}168{+}10$, so $2^{2026} \equiv 2^{10} = 1{,}024 \pmod{13}$. Now $1{,}024 = 13 \times 78 + 10$: the remainder is 10.
2. This is the first form of Fermat's little theorem, $n^{13} \equiv n \pmod{13}$.

**Exercise 8.5 · Divisible by 42.** Prove that for every integer $n$, $n^7 - n$ is divisible by 42.

**Solution.** $42 = 2 \times 3 \times 7$. By Fermat, $n^7 \equiv n \pmod{7}$. Modulo 2: $n^2 \equiv n$, so $n^7 \equiv n$. Modulo 3: $n^3 \equiv n$, so $n^7 = n^3 \times n^3 \times n \equiv n \times n \times n = n^3 \equiv n$. Thus 2, 3 and 7 divide $n^7 - n$; they are pairwise coprime, so (corollary of Gauss's theorem, applied twice) $42 \mid n^7 - n$.

**Exercise 8.6 · Square of a prime number.** Let $p$ be a prime number greater than or equal to 5. Prove that 24 divides $p^2 - 1$.

**Solution.** $p^2 - 1 = (p-1)(p+1)$. Since $p$ is odd, $p-1$ and $p+1$ are two consecutive even integers: one of them is a multiple of 4, so $8 \mid p^2 - 1$. Since $p$ is not a multiple of 3, $p \equiv 1$ or $2 \pmod 3$, so $3 \mid p-1$ or $3 \mid p+1$. 8 and 3 are coprime: $24 \mid p^2 - 1$.

**Exercise 8.7 · Mersenne numbers.**

1. Using the factorisation of $z^m - 1$ (chapter 4), prove that if $n = ab$ with $a \geq 2$ and $b \geq 2$, then $2^a - 1$ divides $2^n - 1$.
2. Deduce that if $2^n - 1$ is prime, then $n$ is prime.
3. Is the converse true? Compute $2^{11} - 1$ and check that 23 divides it.

**Solution.**

1. With $z = 2^a$: $2^n - 1 = z^b - 1 = (z-1)\left(z^{b-1} + \cdots + z + 1\right)$, so $2^a - 1 \mid 2^n - 1$.
2. If $n \geq 2$ is not prime, $n = ab$ as in 1, and $1 < 2^a - 1 < 2^n - 1$ is a divisor of $2^n - 1$, which is therefore not prime. Finally $2^1 - 1 = 1$ is not prime. By contraposition, if $2^n - 1$ is prime, $n$ is prime.
3. No: 11 is prime, but $2^{11} - 1 = 2{,}047 = 23 \times 89$.

## Problem

**Problem · A miniature RSA cipher.** Choose two primes $p = 5$ and $q = 11$, and let $n = pq = 55$ and $\varphi = (p-1)(q-1) = 40$.

1. Choose $e = 3$. Explain why $e$ and $\varphi$ are coprime, then find an integer $d \in \{1, \dots, 39\}$ such that $ed \equiv 1 \pmod{40}$.
2. Encryption: a message is an integer $m \in \{0, \dots, 54\}$; it is encrypted as the remainder $c$ of $m^e$ modulo $n$. Encrypt $m = 8$.
3. Decryption: we take for granted for now that $c^d \equiv m \pmod{n}$. Check with a calculator or Python that $17^{27} \equiv 8 \pmod{55}$.
4. We prove that $m^{ed} \equiv m \pmod{55}$ for every integer $m$.
   (a) Write $ed = 1 + 40k$ with $k$ a natural number.
   (b) If $5 \nmid m$, prove that $m^{40k} \equiv 1 \pmod{5}$; deduce $m^{ed} \equiv m \pmod{5}$. What happens if $5 \mid m$?
   (c) Establish likewise $m^{ed} \equiv m \pmod{11}$, then conclude.
5. Why does the security of this cipher rest on the difficulty of factorising $n$ when $p$ and $q$ are very large?

**Solution · Problem.**

1. $40 = 2^3 \times 5$ and 3 does not divide 40: they are coprime. $3 \times 27 = 81 = 2 \times 40 + 1$: $d = 27$.
2. $8^3 = 512 = 9 \times 55 + 17$: $c = 17$.
3. We do get 8 (for example `pow(17, 27, 55)`).
4. (a) $ed = 81 = 1 + 40 \times 2$: $k = 2$.
   (b) By Fermat, $m^4 \equiv 1 \pmod{5}$, so $m^{40k} = \left(m^4\right)^{10k} \equiv 1$ and $m^{ed} = m \times m^{40k} \equiv m \pmod{5}$. If $5 \mid m$, both sides are congruent to 0.
   (c) Modulo 11: if $11 \nmid m$, $m^{10} \equiv 1$, so $m^{40k} = \left(m^{10}\right)^{4k} \equiv 1$; if $11 \mid m$, both sides are zero modulo 11. Thus 5 and 11 divide $m^{ed} - m$ and are coprime: $55 \mid m^{ed} - m$.
5. Computing $d$ by this method requires $\varphi = (p-1)(q-1)$, hence $p$ and $q$. Knowing only $n$ and $e$, factorising $n$ would be enough to find $d$. For well-chosen primes with several hundred digits, no known method factorises $n$ in a reasonable time on current computers. The security of RSA rests on this assumed difficulty: it has not been proved that every way of decrypting goes through factorising $n$.

**Teacher note · Status of the problem.** RSA is a "possible problem" of the syllabus. Have the encryption and decryption functions written in Python and tested on the 55 possible messages. Question 5 gives an idea of security; it does not call for a proof.